

Frequency-Transformation-Based Co-Designed Lowpass-Single/Multi-Passband-Highpass RF Filters

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Abstract—The theoretical design and practical development of RF analog filtering devices with co-integrated lowpass, single/multi-band bandpass, and highpass transfer functions is reported. For this purpose, two different classes of generalized frequency transformation that convert the equivalent normalized lowpass filter prototype into the desired RF filter with several co-designed filtering actions are proposed. They realize a frequency mapping of the reactance of a normalized unitary capacitor into that of a one-port single/multi-resonance cell with added lowpass and highpass filtering capabilities. For these devised frequency transformations, the theoretical foundations and various illustrative filter examples designed at the ideal-circuit-model level are presented. In addition, higher-selectivity filter architectures based on the generation of additional out-of-band transmission zeros (TZs) by means of cross-coupling techniques are shown. Afterwards, the extension of this design methodology to RF multi-functional filtering components, such as input-reflectionless/absorptive filters based on complementary-diplexer circuit networks and two-way filtering power-distribution circuits, is also demonstrated. Furthermore, two design examples of distributed-element and invert-erless lumped-element RF filters are provided. Besides, for the distributed-element circuit, a proof-of-concept microstrip prototype is manufactured and measured as experimental validation.

Index Terms—Analog filter, absorptive filter, bandpass filter, co-designed circuit, filtering power divider, frequency transformation, highpass filter, lowpass filter, lumped-element filter, microstrip filter, microwave filter, multi-functional circuit, planar filter, reflectionless filter, RF filter, transmission zero (TZ).

I. INTRODUCTION

DURING the last two decades, a considerable research progress on the development of RF/microwave filters with multi-band operational characteristics has been conducted. This has been largely motivated by their potential need for emerging multi-purpose RF-wireless-communications and remote-sensing applications, in which the exploitation of several spectral bands at the same time is advantageous [1]. In this manner, a vast plurality of multi-band bandpass filters for the simultaneous acquisition of multiple desired RF signals corresponding to different services has been reported [2], [3], [4], [5], [6], [7], [8]. The same applies to RF multi-band band-stop or multi-notched-band filters, as a hardware mechanism that may allow the implementation of the required protection for RF systems to unwanted jamming/interference RF signals in highly-congested electromagnetic (EM) environments [9], [10], [11]. Note that although the existing variety of prior-art

RF multi-band filters in terms of underlying RF operational principles and circuit structures is very rich, those approaches using single-to-multi-band frequency transformations deserve special mention [12], [13], [14], [15], [16]. This is owing to their intrinsic simplicity, since they mostly involve the replacement of the basic resonators of a pre-designed bandpass filter counterpart—or of the normalized unitary capacitors of an equivalent normalized lowpass filter prototype for normalized-lowpass-to-multi-band frequency transformations—by sets of resonators to shape the distinct bands. Lately, more-evolved frequency transformations have been devised, such as those in [11] and [17] allowing to embed multiple stopbands inside a wide passband range for the design of RF pre-selection broadband filters with co-integrated multi-interference suppression.

Regarding the design of more-advanced RF filters combining filtering actions of different nature—e.g., lowpass, single/multi-band bandpass, and highpass ones—in composite transfer functions, much less efforts have been made *despite they may be suitable for some advanced RF receivers to separate different frequency regions—e.g., lowpass part associated to a down-converted wide-band signal and other exploitable frequency bands. In tunable implementations, such composite filters may also be used to realize variable-coupling-less adaptive multi-bandstop filters for the dynamic suppression of multiple jamming/interference RF signals with narrow-to-ultra-broad-bandwidth characteristics.* For example, in [18], [19], and [20], RF planar filters with merged lowpass and single/dual-passband functionalities for frequency-static and switchable circuit realizations are described. All of them are based on channelized filter banks, which involve high design complexity in the inter-connection of all channels to properly match all the transmission bands and large circuit size as incorporating more bands needs for additional channels. As an alternative approach, a lowpass-dual-band-bandpass planar filter using metamaterial-inspired cells is proposed in [21]. Nevertheless, it suffers from very-narrow stopband bandwidth above the second passband. In [22], a class of multi-stub-loaded-based double-lowpass/bandpass RF planar filters is presented. Whereas these filters have the merits of abrupt cut-off slopes between transmission and rejection bands owing to multi-TZ creation, they are restricted to such specific transmission-line realizations and to just one passband plus the lowpass one—as the upper passbands that are measured are treated as spurious ones, which are not targeted in the design process. A super-conductor-technology RF filter with high-selectivity composite-lowpass-bandpass transfer function is engineered in [23], but using a two-layer configuration that augments design complexity. Sharp-rejection lowpass-single-passband RF planar filters based on transversal-signal-

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interference principles are suggested in [24]. However, their design procedure fully relies on optimization and its applicability is restricted to filtering responses with one single pass-band. Note that digital-filter-inspired synthesis techniques have also been exploited to realize RF planar filters with transfer functions exhibiting lowpass and bandpass terms as one of the first solutions as in [25], but suffering from low attenuation levels between bands as no inter-band TZs are generated. Hence, it is clear from this description that all these approaches are constrained by insufficient design generality in terms of realizable multi-purpose/co-designed filtering action—none of them use generalized frequency transformations—, circuit structure, and/or applicable RF technology.

This paper presents different types of generalized/multi-purpose frequency transformations for the realization of RF filtering devices with composite-lowpass-single/multi-passband-highpass transfer functions. The conceived frequency transformations can be understood as more-advanced versions of the single-to-multi-band ones presented in the past—e.g., those in [12], [13], [14], [15], and [16]—by also incorporating lowpass-and highpass-type terms. By means of their application over an equivalent normalized lowpass filter prototype to readily obtain the co-designed RF filter, they produce a frequency mapping of the reactance of the normalized unitary capacitor into the one of a constituent one-port multi-resonance cell with co-integrated lowpass and highpass filtering capabilities. In this manner, the coupling circuit topology and pre-synthesized characteristics of the equivalent normalized lowpass filter prototype, such as class of filtering response—e.g., Butterworth-, Chebyshev-, Bessel-, or elliptic-type—, order, and minimum in-band power-matching levels, are directly transferred to all the transmission bands of the converted RF filter. Furthermore, the usefulness of these generalized frequency transformations can be extended well beyond basic RF filters as long as their frequency-selective behavior can be represented through the referred equivalent normalized circuit prototype in the lowpass domain; for example, RF multi-functional filtering components that merge various RF-analog-signal-processing actions into the same RF circuit device for size-reduction and operational-performance advantages, such as filtering power-distribution circuits—e.g., filtering power dividers and couplers—, filtering phase shifters, filtering switches, filtering amplifiers, or even filtering antennas [26], [27], [28], [29]. To the best of the authors' knowledge, a theoretical study as the one of this work about multi-purpose frequency transformations combining lowpass, single/multi-band bandpass, and highpass filtering terms has not been reported in the past. This paper counteracts this lack as a further step for the synthesis of RF filters with co-designed lowpass, single/multi-passband, and highpass terms with regard to the above-referred works in this research area—i.e., [18], [19], [20], [21], [22], [23], [24], and [25]—, which are limited to specific circuit networks, responses—e.g., lowpass-single/multi-passband ones—, and/or RF technologies.

The rest of the manuscript is organized as follows. In Section II, the theoretical foundations of the proposed generalized frequency transformations for co-designed lowpass-single/multi-passband-highpass RF filters, their associated

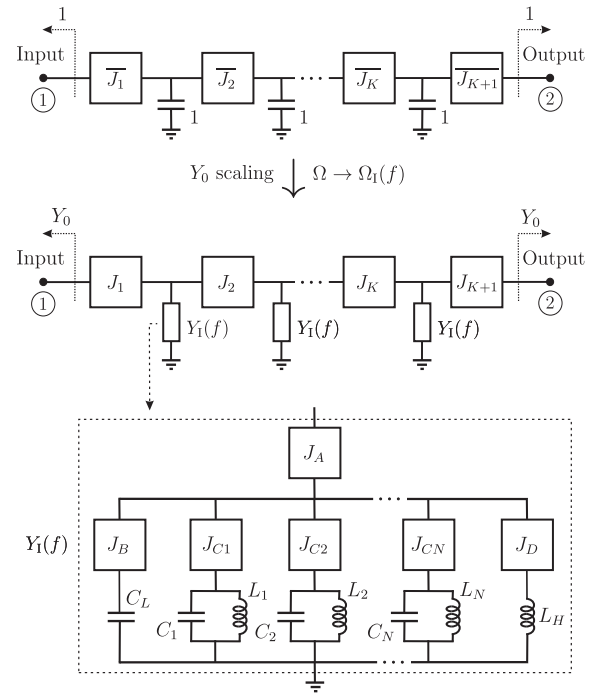


Fig. 1. Circuit detail of the K th-order equivalent normalized-to- Y_0 lowpass filter prototype and the resulting co-designed lowpass-single/multi-passband-highpass filter network after the application of the frequency transformation $\Omega \rightarrow \Omega_1(f)$ (L , C , J , and $\bar{J}$ variables designate inductances, capacitances, admittance-inversion constants, and normalized admittance-inversion constants with regard to the reference admittance Y_0 , respectively).

lumped-element circuit models, theoretical design examples, and techniques for increased-selectivity RF filter design with cross-coupling for extra TZ creation are presented. The applicability of these frequency transformations to more-advanced RF filtering devices, such as RF multi-functional filtering circuits consisting of input-reflectionless filters and two-way filtering power dividers, is also shown. In Section IV, two distributed-element and inverterless lumped-element RF filter designs with composite filtering actions are reported. As experimental validation, a proof-of-concept microstrip prototype of the distributed-element RF filter is built and tested. Finally, the main concluding remarks of this work are set out in Section V.

II. RF FILTERS WITH CO-DESIGNED LOWPASS, SINGLE/MULTI-PASSBAND, AND HIGHPASS RESPONSE

This section reports the basic theoretical foundations of the two conceived classes of composite lowpass-single/multi-passband-highpass RF filter. Specifically, two families of generalized frequency transformation for their analytical design are described, along with illustrative synthesis examples based on their resulting lumped-element circuit networks. Besides, techniques to increase their filtering selectivity by means of the exploitation of additional coupling paths aimed at generating an increased number of out-of-band TZs are also presented.

A. Frequency Transformation Type I

The analytical formula of the first proposed type of generalized normalized-lowpass-to-lowpass-single/multi-passband-highpass frequency transformation, $\Omega \rightarrow \Omega_1(f)$, is as follows:

$$\Omega \rightarrow \Omega_1(f) = \underbrace{\frac{\overline{J}_B^2}{f^2} \alpha}_{\text{Lowpass}} + \underbrace{\sum_{i=1}^N \frac{\overline{J}_{Ci}^2 \beta_i f}{f^2 - f_i^2}}_{N\text{-passband}} - \underbrace{\frac{\overline{J}_D^2}{f}}_{\text{Highpass}} \quad (1)$$

where Ω is the normalized—i.e., lowpass-domain—frequency variable, f is the denormalized—i.e., bandpass-domain—frequency variable, α and γ are frequency-scaling factors for the lowpass and highpass terms, respectively, β_i and f_i are the bandwidth-scaling factor and the center frequency of the i th passband ($i = 1, 2, \dots, N$), and $\overline{J}_A, \overline{J}_B, \overline{J}_{C1}, \overline{J}_{C2}, \dots, \overline{J}_{CN}, \overline{J}_D$ are normalized admittance-inversion constants. The effect of this frequency transformation $\Omega \rightarrow \Omega_1(f)$ on the equivalent lowpass filter prototype normalized with regard to a reference admittance Y_0 for a K th-order circuit architecture is depicted in Fig. 1. As can be seen, the frequency-variation profile of the reactance of the unitary capacitor in the equivalent normalized lowpass filter prototype is converted into the one of a one-port single/multi-resonance cell with simultaneous lowpass and highpass behavior in the transformed co-designed filter. In this manner, the lowpass filtering response of the equivalent normalized lowpass filter prototype is transformed into the desired composite lowpass-single/multi-passband-highpass filtering function. Note that the different terms of the resulting co-designed filtering response are shaped by distinct parts of the transformed lumped-element cell—referred to the admittance Y_0 after admittance scaling—through the action of the corresponding admittance inverters as follows: lowpass term by the capacitances C_L , highpass term by the inductances L_H , and i th passband by the parallel-type resonators L_i, C_i . Moreover, any of these terms can be excluded by disregarding the associated mathematical factor as indicated in the frequency transformation in (1) or, equivalently, by omitting their shaping lumped elements in the converted circuit in Fig. 1. Thus, RF filters with only two or even one filtering actions—e.g., lowpass-multi-passband or just multi-passband ones—can also be designed through (1) after its proper particularization.

By identifying the terms of input admittances of the equivalent normalized lowpass filter network after applying the frequency transformation $\Omega \rightarrow \Omega_1(f)$ and the Y_0 -admittance-scaling operation and the converted co-designed filter circuit, the following design formulas for its lumped elements and admittance-inversion constants can be derived:

$$C_L = \frac{Y_0}{2\pi\alpha} \quad L_H = \frac{1}{2\pi\gamma Y_0} \quad (2)$$

$$L_i = \frac{\beta_i}{2\pi f_i^2 Y_0} \quad C_i = \frac{Y_0}{2\pi\beta_i} \quad (3)$$

$$J_{Ci} = \overline{J}_{Ci} Y_0, \quad i = 1, 2, \dots, N, \quad (4)$$

$$J_A = \overline{J}_A Y_0 \quad J_B = \overline{J}_B Y_0 \quad J_D = \overline{J}_D Y_0 \quad (5)$$

$$J_k = \overline{J}_k Y_0, \quad k = 1, 2, \dots, K + 1. \quad (6)$$

For illustration purposes, the action of the frequency transformation $\Omega \rightarrow \Omega_1(f)$ over a normalized third-order lowpass

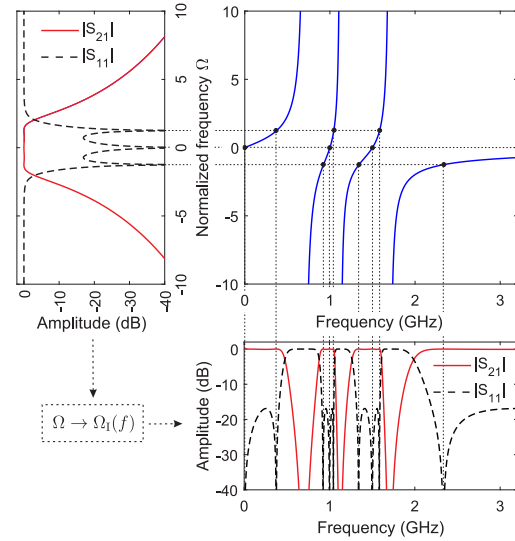


Fig. 2. Effect of the frequency transformation $\Omega \rightarrow \Omega_1(f)$ in terms of power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses: from a third-order equiripple-type equivalent normalized lowpass filter prototype ($K = 3$, $\overline{J}_1 = 1.2$, $\overline{J}_2 = 1.35$, $\overline{J}_3 = 1.35$, and $\overline{J}_4 = 1.2$) to a converted co-designed lowpass-dual-passband-highpass filter for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ (frequency transformation: $N = 2$, $\alpha = 0.4 \text{ GHz}$, $\beta_1 = 0.1 \text{ GHz}$, $\beta_2 = 0.2 \text{ GHz}$, $f_1 = 1.1 \text{ GHz}$, $f_2 = 1.5 \text{ GHz}$, $\gamma = 2 \text{ GHz}$, and $\overline{J}_A = \overline{J}_B = \overline{J}_{C1} = \overline{J}_{C2} = \overline{J}_D = 1$; converted co-designed filter: $C_L = 7.9577 \text{ pF}$, $L_1 = 0.6577 \text{ nH}$, $C_1 = 31.831 \text{ pF}$, $L_2 = 0.7074 \text{ nH}$, $C_2 = 15.9155 \text{ pF}$, and $L_H = 3.9789 \text{ nH}$).

filter network with equiripple-type response to obtain a co-designed lowpass-dual-passband-highpass filter is depicted in Fig. 2. As shown, each term in the composite filtering function is obtained from the mapping of the lowpass filtering response through the corresponding mathematical factor in (1). Note also that inter-band TZs are produced between each two adjacent transmission bands, which come from the transformation of the TZs at $\Omega_{z1,z2} = \pm\infty$ of the normalized lowpass filter.

Additional theoretical synthesis examples of co-designed third-order filtering responses with the frequency transformation $\Omega \rightarrow \Omega_1(f)$ are represented in Fig. 3. They correspond to lowpass-triple-passband-highpass—Fig. 3(a)—, lowpass-single-passband—Fig. 3(b)—, dual-passband-highpass—Fig. 3(c)—, and quadruple-passband—Fig. 3(d)—designs. The values of the design parameters for these synthesis examples—i.e., for the frequency transformation and the converted co-designed filter—are provided in Table I. This validates the generality of this first class of frequency transformation to obtain arbitrary types of co-designed filtering functionalities.

It should be remarked upon that the derivation of the analytical expressions for the cut-off frequencies of the transmission bands of the converted co-designed filter in Fig. 2, $\{f_L < f_{1a} < f_{1b} < f_{2a} < f_{2b} < \dots < f_{Na} < f_{Nb} < f_H\}$, involves solving the following mathematical equation:

$$\Omega_1(f)|_{\{f_L < f_{1a} < f_{1b} < f_{2a} < f_{2b} < \dots < f_{Na} < f_{Nb} < f_H\}} = \pm\Omega_c \quad (7)$$

where Ω_c is the cut-off frequency of the equivalent normalized lowpass filter prototype, f_L and f_H are the cut-off frequencies of the lowpass and highpass terms of the co-designed filtering response, respectively, and f_{ia} and f_{ib} are the lower and upper cut-off frequencies of the i th passband ($i = 1, 2, \dots, N$). In general, this is a complex problem since the design parameters

TABLE I
DESIGN PARAMETERS OF THE CO-DESIGNED THIRD-ORDER ($K = 3$) FILTER EXAMPLES IN FIG. 3

Design example ^(*)	Frequency transformation $\Omega \rightarrow \Omega_1(f)$	Converted co-designed filter
Lowpass-triple-passband-highpass: Fig. 3(a)	$N = 3, \alpha = 0.4 \text{ GHz}, \beta_1 = 0.2 \text{ GHz}, \beta_2 = 0.07 \text{ GHz}, \beta_3 = 0.1 \text{ GHz}, f_1 = 0.9 \text{ GHz}, f_2 = 1.3 \text{ GHz}, f_3 = 1.7 \text{ GHz}, \gamma = 2.5 \text{ GHz}$	$C_L = 7.9577 \text{ pF}, L_1 = 1.9649 \text{ nH}, C_1 = 15.9155 \text{ pF}, L_2 = 0.3296 \text{ nH}, C_2 = 45.4728 \text{ pF}, L_3 = 0.2754 \text{ nH}, C_3 = 31.831 \text{ pF}, L_H = 3.1831 \text{ nH}$
Lowpass-single-passband: Fig. 3(b)	$N = 1, \alpha = 0.4 \text{ GHz}, \beta_1 = 0.15 \text{ GHz}, f_1 = 1.5 \text{ GHz}, \gamma \rightarrow \infty$	$C_L = 7.9577 \text{ pF}, L_1 = 0.5305 \text{ nH}, C_1 = 21.2207 \text{ pF}$
Dual-passband-highpass: Fig. 3(c)	$N = 2, \alpha = 0, \beta_1 = 0.1 \text{ GHz}, \beta_2 = 0.1 \text{ GHz}, f_1 = 0.5 \text{ GHz}, f_2 = 1 \text{ GHz}, \gamma = 2.5 \text{ GHz}$	$L_1 = 3.1831 \text{ nH}, C_1 = 31.831 \text{ pF}, L_2 = 0.7958 \text{ nH}, C_2 = 31.831 \text{ pF}, L_H = 3.1831 \text{ nH}$
Quadruple-passband: Fig. 3(d)	$N = 4, \alpha = 0, \beta_1 = 0.1 \text{ GHz}, \beta_2 = 0.2 \text{ GHz}, \beta_3 = 0.1 \text{ GHz}, \beta_4 = 0.15 \text{ GHz}, f_1 = 0.5 \text{ GHz}, f_2 = 1 \text{ GHz}, f_3 = 1.6 \text{ GHz}, f_4 = 2 \text{ GHz}, \gamma \rightarrow \infty$	$L_1 = 3.1831 \text{ nH}, C_1 = 31.831 \text{ pF}, L_2 = 1.5915 \text{ nH}, C_2 = 15.9155 \text{ pF}, L_3 = 0.3108 \text{ nH}, C_3 = 31.831 \text{ pF}, L_4 = 0.2984 \text{ nH}, C_4 = 21.2207 \text{ pF}$

(*) In all examples, $\bar{J}_1 = 1.2, \bar{J}_2 = 1.35, \bar{J}_3 = 1.35, \bar{J}_4 = 1.2$, and $\bar{J}_A = \bar{J}_B = \bar{J}_{C1} = \bar{J}_{C2} = \dots = \bar{J}_{CN} = \bar{J}_D = 1$.

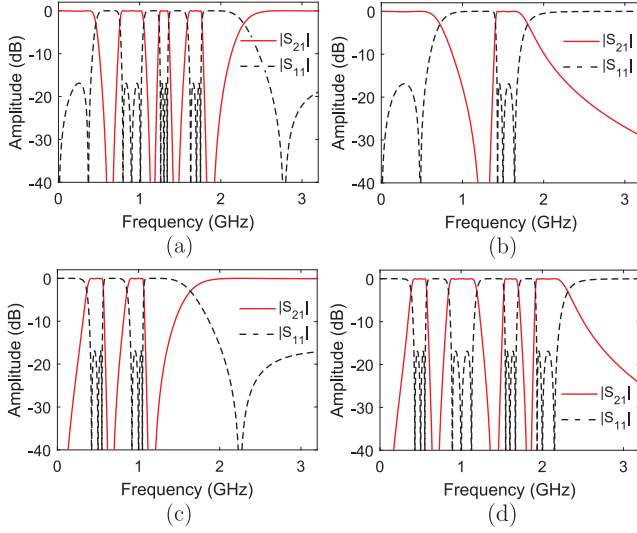


Fig. 3. Theoretical examples of co-designed third-order filters using frequency transformation $\Omega \rightarrow \Omega_1(f)$ in terms of power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ —the design parameters of the frequency transformation and the converted co-designed filter for all the examples are given in Table I. (a) Lowpass-triple-passband-highpass design. (b) Lowpass-single-passband design. (c) Dual-passband-highpass design. (d) Quadruple-passband design.

of all the terms of the co-designed filtering function have an influence on the others. This is easily confirmed for the simple case of a composite lowpass-highpass filtering response—i.e., no passbands are considered—for which the formulas of the cut-off frequencies f_L and f_H turn out to be as follows:

$$f_L, f_H = \frac{\gamma}{2\bar{J}_B^2 \Omega_c} \left(\pm \bar{J}_A^2 + \sqrt{\bar{J}_A^4 + \frac{4\alpha \bar{J}_B^2 \bar{J}_D^2 \Omega_c^2}{\gamma}} \right). \quad (8)$$

Note that as the lowpass-highpass-type response can be regarded as a bandstop one, the above expressions are analogous to those derivable by considering parallel-type resonators in a bandstop-filter architecture with $L_{BS} = C_L/J_B^2$ and $C_{BS} = J_D^2 L_H$ after applying a normalized-lowpass-to-bandstop frequency transformation with a normalized admittance-inversion constant $\bar{J}_A$ and α and γ as the frequency-scaling factors for the lower and upper edges of the shaped stopband.

Finally, it must be highlighted that sharper-rejection co-designed filters through the inclusion of multiple coupling

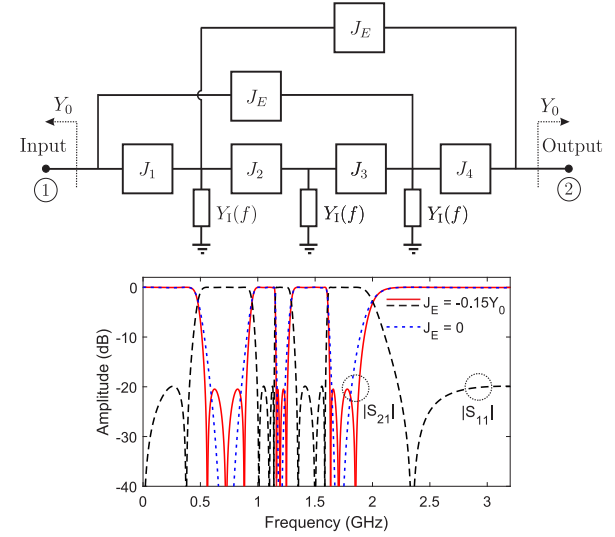


Fig. 4. Theoretical example of higher-selectivity co-designed lowpass-dual-passband-highpass filter in terms of circuit detail and power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$: cross couplings ($J_E = -0.15Y_0$) between the input and output ports and the third and second admittances $Y_1(f)$, respectively—the design parameters are the same as those listed in the caption of Fig. 2. For comparison purposes, the $|S_{21}|$ response without cross couplings (i.e., $J_E = 0$) is also included.

paths for the generation of additional out-of-band TZs can be synthesized. Two examples of such higher-selectivity filter designs for lowpass-dual-passband-highpass and dual-passband-highpass responses in third-order circuit realizations are shown in Figs. 4 and 5, respectively. As can be seen, in the case of Fig. 5, two cross couplings are introduced from the input and output ports to the connecting nodes of the third and second admittances $Y_1(f)$, respectively. On the contrary, in the example of Fig. 4, one additional coupling path is inserted between each pair of contiguous admittances $Y_1(f)$ at specific connecting points inside them. In both cases, more-abrupt cut-off slopes with the creation of extra out-of-band TZs at both sides of all the transmission bands are obtained, while the in-band input-power-matching levels are higher than 20 dB and 17.5 dB in the examples of Figs. 4 and 5, respectively.

B. Frequency Transformations Type II

As the second family of generalized normalized-lowpass-to-lowpass-single/multi-passband-highpass frequency transfor-

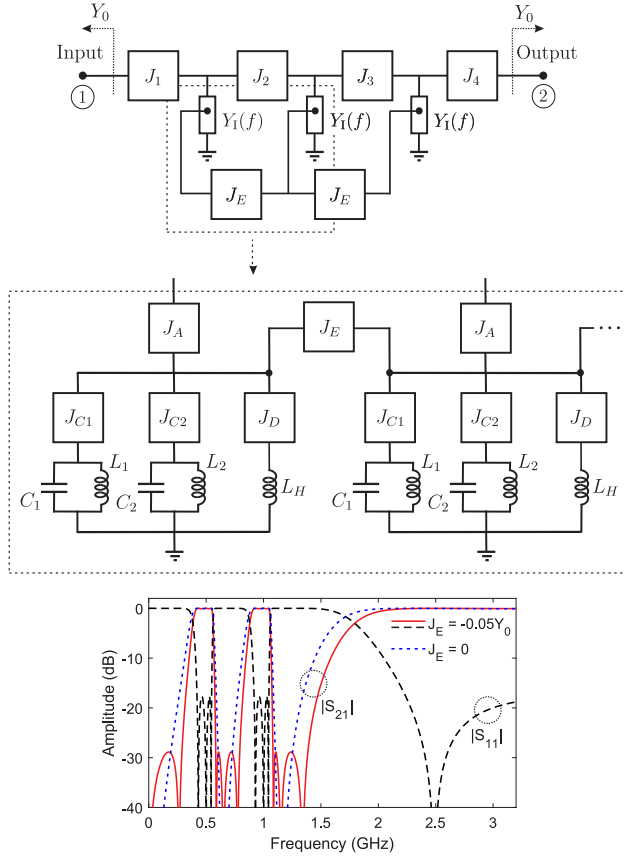


Fig. 5. Theoretical example of higher-selectivity co-designed dual-passband-highpass filter in terms of circuit detail and power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$: additional couplings ($J_E = -0.05Y_0$) between internal interconnection nodes of contiguous admittances $Y_1(f)$ —the design parameters are the same as those listed in Table I for Fig. 3(c). For comparison purposes, the $|S_{21}|$ response without additional couplings (i.e., $J_E = 0$) is also included.

mations, three sub-types—i.e., Sub-Type II-A, Sub-Type II-B, and Sub-Type II-C—are proposed here. Unlike the generalized frequency transformation Type I presented in Section II-A, in which a separated mathematical factor can be identified for the lowpass, single/multi-passband, and highpass terms, these sub-types of frequency transformations exploit the embedding of stopbands within transmission ranges. In this manner, two or even the three of the referred lowpass, single/multi-passband, and highpass filtering actions are generated together by a single mathematical factor. More detail about these three sub-types of generalized frequency transformations is given below.

1) *Sub-Type II-A*: In this generalized frequency transformation, the composite-lowpass-single/multi-passband-highpass filtering response is obtained by splitting an all-pass transmission range by means of several embedded stopbands. Its mathematical expression $\Omega \rightarrow \Omega_{II-A}(f)$ is the following one:

$$\Omega \rightarrow \Omega_{II-A}(f) = \underbrace{\sum_{i=1}^N \frac{\overline{J_{Ai}}^2 \beta_i f}{f^2 - f_i^2}}_{\substack{N\text{-stopband} \equiv \\ \text{Lowpass-(N-1)-} \\ \text{passband-highpass}}} \quad (9)$$

where β_i and f_i are here the bandwidth-scaling factor and the center frequency of the i th inserted stopband ($i =$

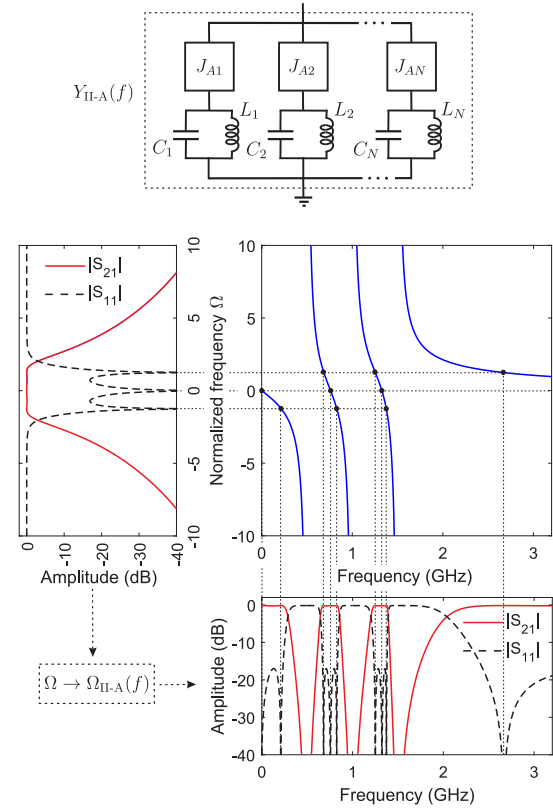


Fig. 6. Effect of the frequency transformation $\Omega \rightarrow \Omega_{II-A}(f)$ in terms of converted one-port single/multi-resonance cell with added lowpass and highpass filtering capabilities after Y_0 scaling for $N - 1$ passbands—i.e., converted admittance $Y_{II-A}(f)$ —and in terms of power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses: from a third-order equiripple-type equivalent normalized lowpass filter prototype ($K = 3$, $\overline{J_1} = 1.2$, $\overline{J_2} = 1.35$, $\overline{J_3} = 1.35$, and $\overline{J_4} = 1.2$) to a converted co-designed lowpass-dual-passband-highpass filter for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ (frequency transformation: $N = 3$, $\beta_1 = 0.15 \text{ GHz}$, $\beta_2 = 0.1 \text{ GHz}$, $\beta_3 = 0.1 \text{ GHz}$, $f_1 = 0.5 \text{ GHz}$, $f_2 = 1.1 \text{ GHz}$, $f_3 = 1.5 \text{ GHz}$, $\overline{J_{A1}} = \overline{J_{A3}} = 3$, and $\overline{J_{A2}} = 2.5$; converted co-designed filter: $L_1 = 3.1831 \text{ nH}$, $C_1 = 31.831 \text{ pF}$, $L_2 = 1.1937 \text{ nH}$, $C_2 = 21.2207 \text{ pF}$, $L_3 = 0.3537 \text{ nH}$, and $C_3 = 31.831 \text{ pF}$).

$1, 2, \dots, N$). Note that this frequency transformation can be equivalently viewed as a normalized-lowpass-to-single/multi-stopband one, in which the embedding of N stopbands leads to the creation of $N - 1$ passbands plus the lowpass and highpass terms. Thus, the particular case $N = 1$ corresponds to a co-designed lowpass-highpass filtering action without passbands.

Fig. 6 depicts the circuit detail of the one-port single/multi-resonance cell with added lowpass and highpass filtering behavior and associated admittance $Y_{II-A}(f)$ obtained after applying the generalized frequency transformation $\Omega \rightarrow \Omega_{II-A}(f)$ and Y_0 admittance scaling, whose circuit element values can be calculated by means of (10)–(12). The effect of such frequency transformation over a normalized third-order lowpass filter network with equiripple-type response to design a lowpass-dual-passband-highpass filter is also illustrated in this figure.

$$L_i = \frac{\beta_i}{2\pi f_i^2 Y_0} \quad C_i = \frac{Y_0}{2\pi \beta_i}, \quad (10)$$

$$J_{Ai} = \overline{J_{Ai}} Y_0, \quad i = 1, 2, \dots, N, \quad (11)$$

$$J_k = \overline{J_k} Y_0, \quad k = 1, 2, \dots, K + 1. \quad (12)$$

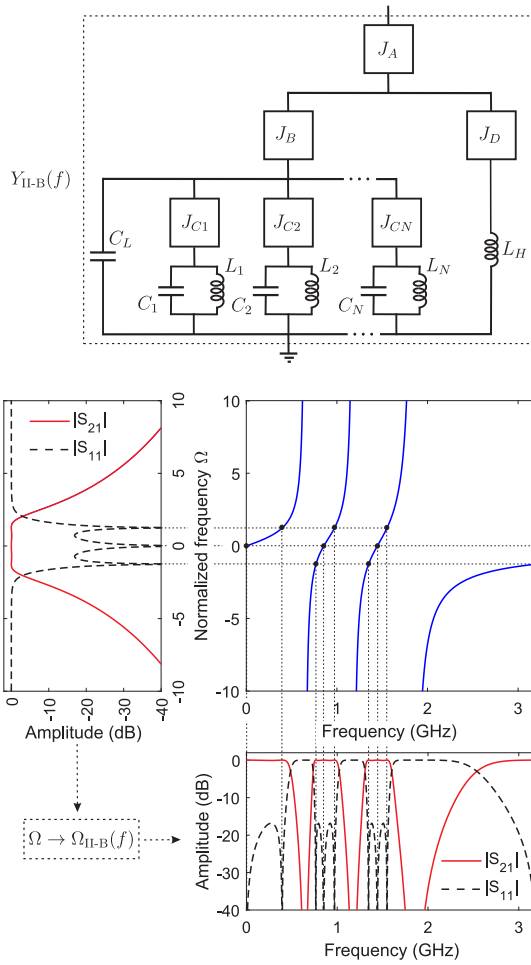


Fig. 7. Effect of the frequency transformation $\Omega \rightarrow \Omega_{II-B}(f)$ in terms of converted one-port single/multi-resonance cell with added lowpass and highpass filtering capabilities after Z_0 scaling for N passbands—i.e., converted admittance $Y_{II-B}(f)$ —and in terms of power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses: from a third-order equiripple-type equivalent normalized lowpass filter prototype ($K = 3$, $\overline{J}_1 = 1.2$, $\overline{J}_2 = 1.35$, $\overline{J}_3 = 1.35$, and $\overline{J}_4 = 1.2$) to a converted co-designed lowpass-dual-passband-highpass filter for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ (frequency transformation: $N = 2$, $\alpha = 0.85$ GHz, $\beta_1 = 0.1$ GHz, $\beta_2 = 0.1$ GHz, $\gamma = 3$ GHz, $f_1 = 0.7$ GHz, $f_2 = 1.3$ GHz, $\overline{J}_A = \overline{J}_B = \overline{J}_D = 1$, and $\overline{J}_{C1} = \overline{J}_{C2} = 2$; converted co-designed filter: $C_L = 3.9789$ pF, $L_1 = 1.624$ nH, $C_1 = 31.831$ pF, $L_2 = 0.4709$ nH, $C_2 = 31.831$ pF, and $L_H = 2.6526$ nH).

2) *Sub-Type II-B*: In this generalized frequency transformation, one or multiple stopbands are inserted within the lowpass transmission range of a composite-lowpass-highpass response. Its analytical formula $\Omega \rightarrow \Omega_{II-B}(f)$ is provided below:

$$\Omega \rightarrow \Omega_{II-B}(f) = \frac{\overline{J}_A^2}{\frac{\overline{J}_B^2}{\frac{f}{\alpha} - \sum_{i=1}^N \frac{\overline{J}_{Ci}^2 \beta_i f}{f^2 - f_i^2}} - \frac{\overline{J}_D^2}{\gamma}}. \quad (13)$$

Lowpass/ N -stopband $\equiv$ Lowpass/ N -passband

It should be remarked upon that in this frequency transformation, whereas β_i is the bandwidth-scaling factor of the i th inserted stopband ($i = 1, 2, \dots, N$), f_i is related—but it does not exactly correspond—with its center frequency since

its location in the co-designed filtering action also depends on the highpass factor in (13). However, if such highpass term is omitted, then f_i becomes the center frequency of the i th stopband in the resulting co-designed lowpass-single/multi-passband filtering action. Note also that, in this case, N stopbands create N passbands in the composite transfer function.

The circuit detail of the one-port single/multi-resonance cell with co-integrated lowpass and highpass filtering capabilities and associated admittance $Y_{II-B}(f)$ resulting from the generalized frequency transformation $\Omega \rightarrow \Omega_{II-B}(f)$ and Y_0 admittance scaling is shown in Fig. 7. Its effect over a normalized third-order lowpass filter network with equiripple-type response to design a lowpass-dual-passband-highpass filter is also exemplified. The formulas to compute the lumped-element values of the converted admittance $Y_{II-B}(f)$ are given in (14)–(18).

$$C_L = \frac{Y_0}{2\pi\alpha} \quad L_H = \frac{1}{2\pi\gamma Y_0} \quad (14)$$

$$L_i = \frac{\beta_i}{2\pi f_i^2 Y_0} \quad C_i = \frac{Y_0}{2\pi\beta_i} \quad (15)$$

$$J_{Ci} = \overline{J}_{Ci} Y_0, \quad i = 1, 2, \dots, N, \quad (16)$$

$$J_A = \overline{J}_A Y_0 \quad J_B = \overline{J}_B Y_0 \quad J_D = \overline{J}_D Y_0 \quad (17)$$

$$J_k = \overline{J}_k Y_0, \quad k = 1, 2, \dots, K + 1. \quad (18)$$

3) *Sub-Type II-C*: This generalized frequency transformation is the counterpart of the previous one, but with the stopband(s) being embedded in the highpass region instead of in the lowpass part. Its mathematical formula $\Omega \rightarrow \Omega_{II-C}(f)$ is

$$\Omega \rightarrow \Omega_{II-C}(f) = \frac{\overline{J}_A^2}{\underbrace{\frac{\overline{J}_D^2}{\alpha}}_{\text{Lowpass}} - \underbrace{\frac{\gamma}{f} + \sum_{i=1}^N \frac{\overline{J}_{Ci}^2 \beta_i f}{f^2 - f_i^2}}_{\text{Highpass}/N\text{-stopband} \equiv \text{Highpass}/N\text{-passband}}} \quad (19)$$

Analogous considerations to those indicated for the frequency transformation $\Omega \rightarrow \Omega_{II-B}(f)$ can be made in this case. Moreover, as for the previous frequency transformations, the circuit detail of the converted admittance $Y_{II-C}(f)$, the effect of the frequency transformation $\Omega \rightarrow \Omega_{II-C}(f)$ for a particular design example, and the design equations for its lumped elements are provided in Fig. 19 and (20)–(24), respectively.

$$C_L = \frac{Y_0}{2\pi\alpha} \quad L_H = \frac{1}{2\pi\gamma Y_0} \quad (20)$$

$$L_i = \frac{\beta_i}{2\pi f_i^2 Y_0} \quad C_i = \frac{Y_0}{2\pi\beta_i} \quad (21)$$

$$J_{Ci} = \overline{J}_{Ci} Y_0, \quad i = 1, 2, \dots, N, \quad (22)$$

$$J_A = \overline{J}_A Y_0 \quad J_B = \overline{J}_B Y_0 \quad J_D = \overline{J}_D Y_0 \quad (23)$$

$$J_k = \overline{J}_k Y_0, \quad k = 1, 2, \dots, K + 1. \quad (24)$$

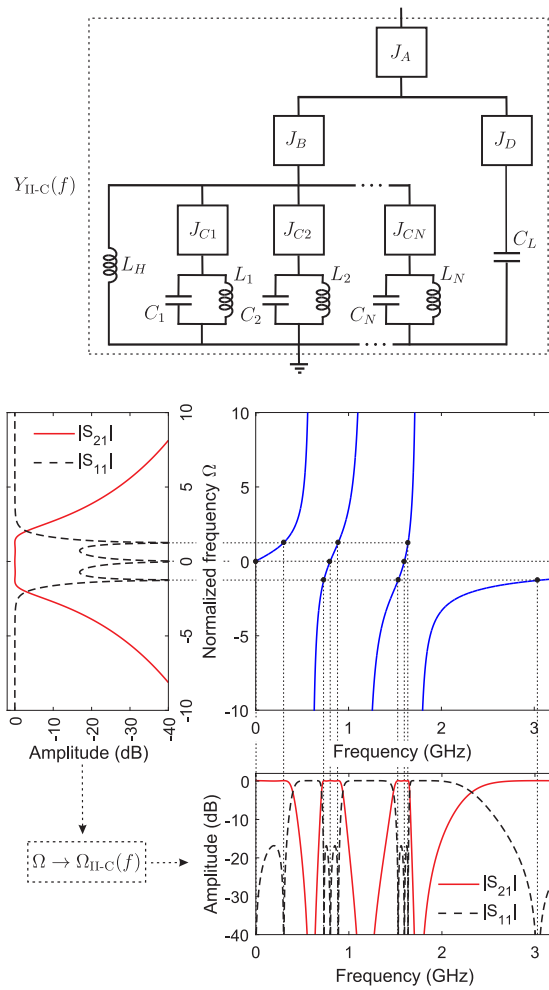


Fig. 8. Effect of the frequency transformation $\Omega \rightarrow \Omega_{II-C}(f)$ in terms of converted one-port single/multi-resonance cell with added lowpass and highpass filtering capabilities after Z_0 scaling for N passbands—i.e., converted admittance $Y_{II-C}(f)$ —and in terms of power transmission ($|S_{21}|$) and reflection ($|S_{11}|$) responses: from a third-order equiripple-type equivalent normalized lowpass filter prototype ($K = 3$, $J_1 = 1.2$, $J_2 = 1.35$, $J_3 = 1.35$, and $J_4 = 1.2$) to a converted co-designed lowpass-dual-passband-highpass filter for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ (frequency transformation: $N = 2$, $\alpha = 0.3 \text{ GHz}$, $\beta_1 = 0.1 \text{ GHz}$, $\beta_2 = 0.1 \text{ GHz}$, $\gamma = 1.8 \text{ GHz}$, $f_1 = 1 \text{ GHz}$, $f_2 = 1.7 \text{ GHz}$, $J_A = J_B = J_D = 1$, $J_{C1} = 3$, and $J_{C2} = 2$; converted co-designed filter: $C_L = 10.6103 \text{ pF}$, $L_1 = 0.7958 \text{ nH}$, $C_1 = 31.831 \text{ pF}$, $L_2 = 0.2754 \text{ nH}$, $C_2 = 31.831 \text{ pF}$, and $L_H = 0.2984 \text{ nH}$).

4) *Additional Considerations:* In relation with the three sub-types of generalized frequency transformations presented in this section, as well as the one described in Section II-A, the following additional aspects must be taken into account:

- The selection of the spectral parameters—i.e., center frequencies and bandwidth-scaling factors—in the suggested generalized frequency transformations must be done in accordance to their meaning. For example, such parameters must be chosen so that the embedded stopbands are contained within the lowpass region in the Sub-Type II-B or inside the highpass one in the Sub-Type II-C. Similarly, the passband(s) in the Type I must be located within the stopband region between the lowpass and highpass terms. Besides, note that similar design-flexibility properties as those exemplified in Fig. 3 for the frequency transforma-

tion Type I are feasible for the frequency transformations Sub-Type II-A, Sub-Type II-B—except single/multi-passband-highpass responses—, and Sub-Type II-C—except lowpass-single/multi-passband responses.

- Equation (7) also applies to the generalized frequency transformations Sub-Type II-A, Sub-Type II-B, and Sub-Type II-C with $\Omega_{II-A}(f)$, $\Omega_{II-C}(f)$, and $\Omega_{II-C}(f)$, respectively, for the analytical derivation of the expressions of the cut-off frequencies of the transmission bands, which is a complex mathematical problem except for simple cases.
- Whereas the generalized frequency transformation Sub-Type II-A inherently introduces lowpass and highpass filtering actions in the co-designed transfer function, additional circuit elements can be added to the converted circuit to eliminate any of them. For example, input/output lowpass filtering cells may be added to suppress the highpass range, so that an overall lowpass-single/multi-passband filtering response is obtained. This technique is exemplified in Section IV-A by means of the design of a distributed-element lowpass-dual-band-bandpass filter, in which transversal-signal-interference lowpass-type filtering cells are inserted to mitigate the highpass range—that, on the other hand, gets distorted by the spurious resonances and frequency-dependent behavior of the resonators and admittance inverters, respectively, in such RF transmission-line-based circuit realization.
- For a certain set of specifications that define the targeted attenuation mask, all the presented types of generalized frequency transformations allow to theoretically obtain lowpass-single/multi-passband-highpass filtering responses with comparable characteristics. Thus, the election of one or another of them is mostly determined by practical aspects related with the difficulties involved in realizing such transfer function in the adopted RF technology. For example, broad spectral spacings between the multiple passbands are more feasible in practice with the frequency transformation Type I, whereas narrower interband separations are better suited for the frequency transformations Sub-Type II-A, Sub-Type II-B, and Sub-Type II-C. This is due to the difficulties involved on producing very-strong couplings in RF technologies, as usually required to create wide stopbands—i.e., broad spacings between passbands—in the frequency transformations Sub-Type II-A, Sub-Type II-B, and Sub-Type II-C.
- Similar techniques based on the exploitation of additional coupling paths as those expounded in Section II-A can be applied to the generalized frequency transformations Sub-Type II-A, Sub-Type II-B, and Sub-Type II-C for sharper-rejection co-designed filters with extra out-of-band TZs.

III. APPLICATION TO RF MULTI-FUNCTIONAL FILTERING COMPONENTS

Since the engineered classes of composite-lowpass-single/multi-passband-highpass RF filter are based on generalized frequency transformations, the same design philosophy can be extended to RF multi-functional filtering components that co-integrate various RF-analog-signal-processing actions in the

same circuit. This just requires starting from the representation of the RF multi-functional filtering device in terms of its equivalent normalized lowpass prototype, from which the final lumped-element circuit model and associated response can be obtained after applying the corresponding frequency transformation as detailed in Section II. For illustration purposes, such capability is demonstrated here for a complementary-diplexer-based input-reflectionless filter and a filtering power divider. Note also that although a specific type of the proposed frequency transformations is selected in all these examples, the results are extensible to any of the suggested frequency transformations and even to enhanced-selectivity designs by using the cross-coupling methods expounded in Section II-A.

A. Input-Reflectionless Filter

Whereas most of prior-art RF filters exhibit a reflective-type behavior, the design of reflectionless/absorptive filters is lately attracting a large research interest. By including lossy elements in their circuit networks, reflectionless filters dissipate inside themselves the out-of-band/non-transmitted RF-input-signal energy instead of reflecting it back to the source. Hence, they are less harmful than their reflective counterparts for preceding active stages in a full RF front-end chain, such as mixers of frequency-conversion stages and amplifiers [30]. Moreover, as no RF-power reflections are created by these RF filters, the use of inter-stage isolators or attenuators—which may increase overall size/volume, DC-power consumption for active-circuit realizations, or in-band insertion-loss levels—can be avoided.

In the past literature, several methods and circuit topologies to design reflectionless filters have been reported [31], [32]. One of them consists of the use of complementary networks, in which a main channel serves as an approximation to the overall filtering response to be attained, whereas the remaining ones—one or several—are loaded by resistors at their output ports and they jointly implement a complementary frequency response with regard to the one of the main channel. Thus, the out-of-band RF-signal-power reflections coming from the main channel are fully dissipated by the loading resistor(s) of the other channel(s), so that the total filter becomes fully absorptive at its input terminal. Design techniques for reflectionless bandpass filters with Butterworth, Chebyshev, and quasi-elliptic-type responses and multi-passband filters—e.g., split-type ones—have been detailed [33], [34], [35], [36]. However, reflectionless filters with co-designed lowpass-single/multi-passband-highpass transfer functions have not been discussed.

The above-referred limitation is counteracted here by means of the generalized frequency transformations proposed in this work. Taking a complementary-diplexer network and the first class of frequency transformation in Section II-A—i.e., Type I—as an example, the synthesis of a fourth-order input-reflectionless filter with lowpass-triple-passband response has been carried out. To this aim, the normalized equivalent lowpass representation detailed in [33] is adopted, to which the referred frequency transformation is then applied. The resulting lumped-element circuit and associated transfer function for the co-designed fourth-order lowpass-triple-passband filter after such frequency transformation are drawn in Fig. 9. The

derived values for the design parameters of the converted filter are summarized in its caption, for which the design equations (2)–(5) along with those extracted from [33] that are provided below for the admittance-inversion constants were employed:

$$J_1 = \frac{\overline{J_{13}}}{\overline{J_{14}}} \sqrt{\frac{1 + \sqrt{2}}{\sqrt{2}}} Y_0 \quad J_2 = \left(\frac{\overline{J_{13}}}{\overline{J_{14}}} \right)^2 \sqrt{2\sqrt{2}} Y_0 \quad (25)$$

$$J_3 = \left(\frac{\overline{J_{13}}}{\overline{J_{14}}} \right)^2 2Y_0 \quad J_4 = \left(\frac{\overline{J_{13}}}{\overline{J_{14}}} \right)^2 \sqrt{2(4 + 3\sqrt{2})} Y_0 \quad (26)$$

$$J_5 = \frac{\overline{J_{13}}}{\overline{J_{14}}} \sqrt{4 + 2\sqrt{2}} Y_0 \quad J_6 = \frac{\overline{J_8 J_{12}}}{\overline{J_{10} J_{14}}} Y_0 \quad (27)$$

$$J_7 = \frac{\overline{J_8 J_{12} J_{13}}}{(\overline{J_{14}})^2 \overline{J_{10}}} 2Y_0 \quad J_8 = \overline{J_8} Y_0 \quad (28)$$

$$J_9 = \frac{\overline{J_{10} J_{13}}}{\overline{J_{12}}} \sqrt{\frac{3 + 2\sqrt{2}}{\sqrt{2}}} Y_0 \quad J_{10} = \overline{J_{10}} Y_0 \quad (29)$$

$$J_{11} = \frac{\overline{J_{12} J_{13}}}{(\overline{J_{14}})^2} \sqrt{2\sqrt{2}} Y_0 \quad J_{12} = \overline{J_{12}} Y_0 \quad (30)$$

$$J_{13} = \overline{J_{13}} Y_0 \quad J_{14} = \overline{J_{14}} Y_0 \quad (31)$$

where $\overline{J_8}$, $\overline{J_{10}}$, $\overline{J_{12}}$, $\overline{J_{13}}$, and $\overline{J_{14}}$ are variables of freedom in the sense of attaining the input-reflectionless behavior and only $\overline{J_{13}}$ and $\overline{J_{14}}$ have an influence on the total transfer function. As observed in Fig. 9, the expected lowpass-triple-passband response and perfect RF-input-absorptive behavior are obtained.

B. Filtering Power Divider

One of the main and first exponents of co-designed RF filtering components with multiple functionalities are power-distribution networks with simultaneous filtering capabilities, such as filtering power dividers and combiners. This is due to the simplicity of their design, which mostly consists of the co-integration of the filtering units in the power-divider/combiner branches. Despite their realization is relatively straightforward for the case of symmetrical-power-division/combination actions, some aspects need to be considered in asymmetrical-power-distribution implementations—e.g., differences between the frequency-variation profiles of the power transmission responses from the input port to the different output terminals—and designs with isolated output accesses—e.g., location of the isolation resistors—, such as in Wilkinson-type schemes.

As of today, various filtering power dividers with frequency-static single- and multi-band bandpass responses using different design techniques—e.g., co-integration of transversal-signal-interference cells or coupled-resonator networks in the power-divider arms—have been reported [37], [38]. The incorporation of frequency reconfiguration in such co-designed RF filtering power dividers has also been addressed as in [39] and [40], as well as the co-integration of additional operational features, such as the previously-referred reflectionless/absorptive behavior for one or all ports [41], [42]. Nevertheless, the

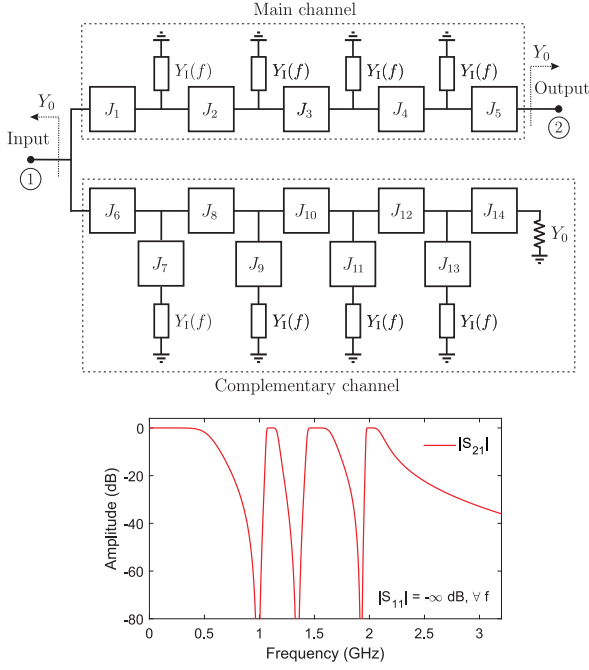


Fig. 9. Theoretical example of complementary-diplexer-based input-reflectionless co-designed fourth-order lowpass-triple-passband filter in terms of circuit detail and power transmission ($|S_{21}|$) response for a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ (frequency transformation Type I: $N = 3$, $\alpha = 0.6$ GHz, $\beta_1 = 0.1$ GHz, $\beta_2 = 0.2$ GHz, $\beta_3 = 0.1$ GHz, $f_1 = 1.1$ GHz, $f_2 = 1.5$ GHz, $f_3 = 2$ GHz, $\gamma \rightarrow \infty$, and $\bar{J}_A = \bar{J}_B = \bar{J}_{C1} = \bar{J}_{C2} = \bar{J}_{C3} = 1$; converted co-designed filter: $C_L = 5.3052$ pF, $L_1 = 0.6577$ nH, $C_1 = 31.831$ pF, $L_2 = 0.7074$ nH, $C_2 = 15.9155$ pF, $L_3 = 0.1989$ nH, $C_3 = 31.831$ pF, $J_1 = 0.8088Y_0$, $J_2 = 0.6444Y_0$, $J_3 = 0.7663Y_0$, $J_4 = 1.5557Y_0$, $J_5 = 1.6175Y_0$, $J_6 = Y_0$, $J_7 = 1.238Y_0$, $J_8 = Y_0$, $J_9 = 1.2566Y_0$, $J_{10} = Y_0$, $J_{11} = 1.041Y_0$, $J_{12} = Y_0$, $J_{13} = 0.619Y_0$, and $J_{14} = Y_0$).

design of filtering power dividers with simultaneous lowpass-single/multi-passband-highpass behavior remains unexplored.

As an attempt to fill the aforementioned void, the theoretical design of a two-way filtering power divider with third-order triple-passband/highpass response for both symmetrical and asymmetrical power-division actions has been conducted. Specifically, the first type of generalized frequency transformation presented in Section II-A—i.e., Type I—has been again selected for demonstrative purposes. The resulting lumped-element-based circuit for the filtering power divider after applying the referred frequency transformation is depicted in Fig. 10, along with the power transmission, reflection, and output-isolation parameters for both design examples. The values for the design parameters for these two examples are provided in the caption, in which a power-division ratio of $k^2:1$ —i.e., $P_3 = k^2P_2$ in Fig. 10 where P_1 is the input power at port 1 and P_2 and P_3 are the output powers at ports 2 and 3, respectively—with $k = \sqrt{2}$ was chosen for the asymmetrical design. The design formulas (2)–(5) were utilized for their calculation, along with the ones detailed below for the isolation resistor and the admittance-inversion constants in the filtering-power-divider branches:

$$J_1 = \bar{J}_1 \sqrt{\frac{1}{k(1+k^2)}} Y_0 \quad J_2 = J_6 = \bar{J}_2 Y_0 \quad (32)$$

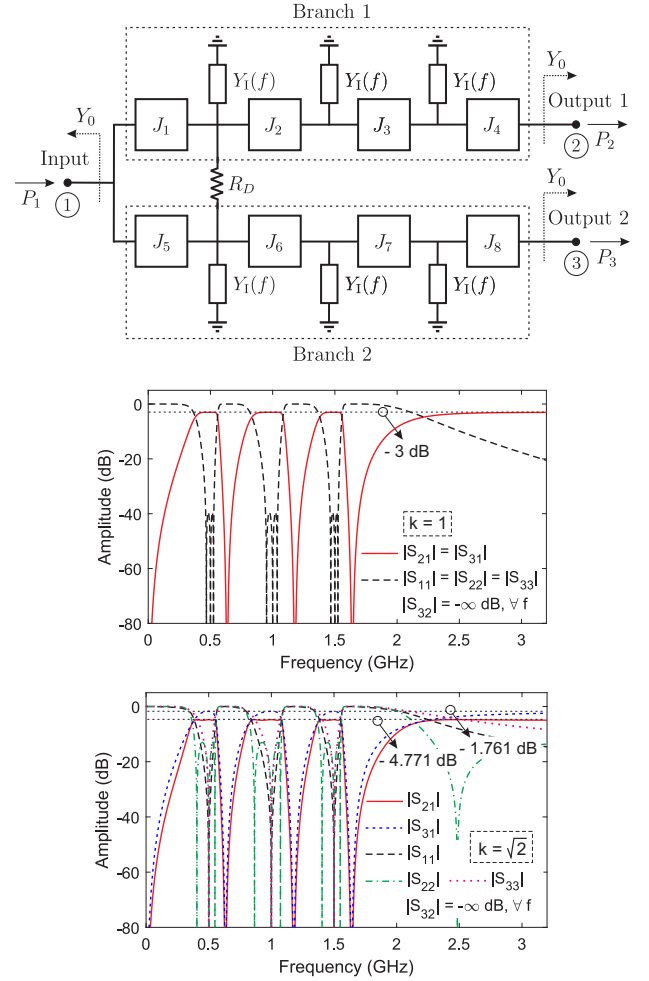


Fig. 10. Theoretical examples of co-designed third-order triple-passband-highpass filtering power divider in terms of circuit detail and power transmission ($|S_{21}|$ and $|S_{31}|$), input/output-reflection ($|S_{11}|$, $|S_{22}|$, and $|S_{33}|$), and output-isolation ($|S_{32}|$) responses for equal ($k = 1$) and unequal ($k = \sqrt{2}$) power-division factors and a reference admittance $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$ (frequency transformation Type I: $N = 3$, $\alpha = 0$, $\beta_1 = 0.1$ GHz, $\beta_2 = 0.15$ GHz, $\beta_3 = 0.1$ GHz, $f_1 = 0.5$ GHz, $f_2 = 1.1$ GHz, $f_3 = 1.5$ GHz, $\gamma = 2.8$ GHz, and $\bar{J}_A = \bar{J}_B = \bar{J}_{C1} = \bar{J}_{C2} = \bar{J}_{C3} = 1$; converted co-designed filtering power divider: $L_1 = 3.1831$ nH, $C_1 = 31.831$ pF, $L_2 = 1.1937$ nH, $C_2 = 21.2207$ pF, $L_3 = 0.3537$ nH, $C_3 = 31.831$ pF, and $L_H = 0.2984$ nH; $\bar{J}_1 = \bar{J}_2 = \bar{J}_3 = \bar{J}_4 = 1.35$; $k = 1$: $J_1 = 0.9546Y_0$, $J_2 = J_6 = 1.35Y_0$, $J_3 = J_7 = 1.35Y_0$, $J_4 = 1.35Y_0$, $J_5 = 0.9546Y_0$, $J_8 = 1.35Y_0$, and $R_D = 1.0974Z_0$; $k = \sqrt{2}$: $J_1 = 0.6554Y_0$, $J_2 = J_6 = 1.35Y_0$, $J_3 = J_7 = 1.35Y_0$, $J_4 = 1.1352Y_0$, $J_5 = 1.3108Y_0$, $J_8 = 1.6054Y_0$, and $R_D = 1.164Z_0$).

$$J_3 = J_7 = \bar{J}_3 Y_0 \quad J_4 = \bar{J}_4 \sqrt{\frac{1}{k}} Y_0 \quad (33)$$

$$J_5 = \bar{J}_1 \sqrt{\frac{k^3}{1+k^2}} Y_0 \quad J_8 = \bar{J}_4 \sqrt{k} Y_0 \quad (34)$$

$$R_D = \left(\frac{\bar{J}_3}{\bar{J}_2 \bar{J}_4} \right)^2 \left(\frac{1+k^2}{k} \right) Z_0 \quad (35)$$

where $\bar{J}_k$ ($k = 1, 2, \dots, K+1$) are the normalized inversion constants of the K th-order equivalent normalized lowpass filter prototype associated to the filtering action to be incorporated in the power divider. As can be seen in Fig. 10, the

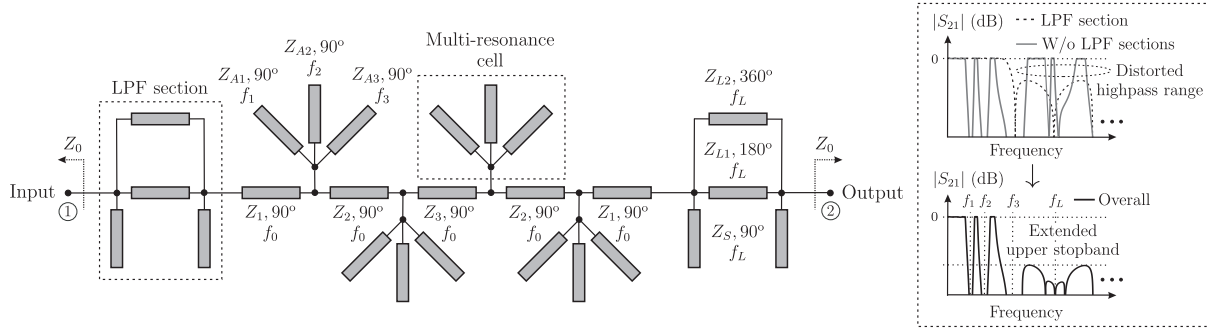


Fig. 11. Circuit detail and conceptual operational principle of the designed fourth-order lowpass-dual-passband RF filter based on the generalized frequency transformation Sub-Type II-A and input/output lowpass filtering (LPF) sections (Z variables denote characteristic impedances and the values of the electrical lengths of the transmission-line segments are given at the indicated frequencies f_0, f_1, f_2, f_3 , and f_L).

intended power-division/composite-triple-passband-highpass-filtering functions are obtained in both cases, along with high in-band input/output-power-matching levels and all-frequency perfect isolation between the output terminals. Note that spectral asymmetry between branches in terms of bandwidths for the transmission bands and input/output-power-matching levels is observed in this unequal-power-division design example, which may be compensated if needed by optimization.

IV. DESIGN EXAMPLES AND EXPERIMENTAL RESULTS

In order to demonstrate the usefulness of the suggested generalized frequency transformations for the realization of multi-purpose filters with co-designed lowpass-single/multi-band-bandpass-highpass responses in specific RF technologies, two design examples are presented here. They correspond to a fourth-order lowpass-dual-passband microstrip filter and an inverterless third-order lowpass-dual-passband lumped-element filter for a reference impedance $Z_0 = 50 \Omega$.

For the distributed-element RF filter, a proof-of-concept microstrip prototype has been constructed and measured as further practical validation. In the experimental development of this RF filter, a Rogers RO4003C planar substrate with the following parameters was used: relative dielectric constant $\epsilon_r = 3.55$, dielectric thickness $h = 0.813$ mm, metal thickness $t = 18 \mu\text{m}$, and dielectric loss tangent $\tan(\delta_D) = 0.0027$. The EM simulations were carried out by means of the commercial software package Ansys HFSS, whereas the measurements were taken through an Agilent N5230 vector network analyzer. The description of the design process and the theoretical responses of the two designed RF filters, as well as the EM-simulated and measured results of the manufactured microstrip prototype associated to the first of them, are provided below.

A. Design I: Distributed-Element Filter

The first validation example consists of an RF filter with co-integrated fourth-order lowpass and dual-passband filtering actions developed in microstrip technology. Its ideal transmission-line-based circuit scheme and conceptual operational principle is detailed in Fig. 11. Owing to design simplicity for this technology, the generalized frequency transformation Sub-Type II-A was adopted in this example. Specifically, the steps described below were followed for its design process:

- 1) Application of the generalized frequency transformation Sub-Type II-A given in (9) over a pre-designed equivalent normalized lowpass filter prototype with the intended in-band ripple level—i.e., 0.25 dB for this design example—and Z_0 impedance scaling. The ripple-level-referred cut-off frequency of the lowpass transmission range was fixed as 0.85 GHz, whereas the center frequencies of the first and second passbands were selected as 1.4 and 2.2 GHz with absolute bandwidths of 130 and 200 MHz referred to the ripple level, respectively. Such specifications were accomplished after the numerical inspection of (7) with (9) and the composite filtering response of the converted co-designed filter, whose lumped-element and admittance-inversion-constant values were obtained with (10)–(12).
- 2) Realization of the lumped-element resonators plus their adjacent admittance inverters in the converted multi-resonant cell as quarter-wavelength open-ended transmission-line segments and the admittance inverters of the direct input-output path as quarter-wavelength transmission-line segments. To this aim, well-known RF/microwave filter design techniques establishing the referred equivalences between these lumped-circuit elements and their transmission-line-based counterparts were employed [43], [44].
- 3) Separate design of the input/output sharp-rejection lowpass-type filtering cells based on the transversal-signal-interference approach presented in [45] to remove the undesired highpass-type transmission range in the previous circuit part, which gets also distorted by the frequency-periodic behavior of the transmission-line resonators—i.e., spurious resonances—and admittance inverters.
- 4) In-series cascade connection of the designed input/output lowpass filtering sections to shape the overall RF filter as shown in Fig. 11 and fine optimization to maximize the in-band power-matching levels for its lowpass and dual-passband transmission regions along with the power-rejection levels and bandwidth of its upper-stopband range.
- 5) Dimensioning of the filter layout with its final EM optimization using commercial computer-aided-design tools.

The power transmission and input-reflection responses of the theoretically-designed overall filter with composite lowpass and dual-passband response given in Fig. 11 are depicted in Fig. 12. The values for the design parameters are listed in its caption. For comparison purposes, the transfer functions of

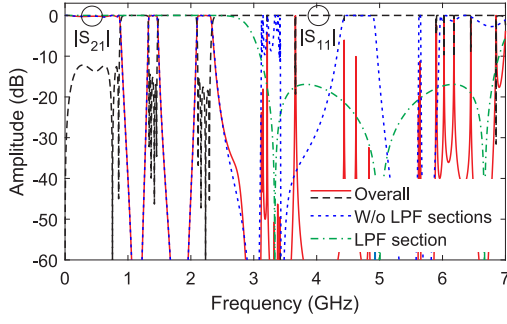


Fig. 12. Power transmission ($|S_{21}|$) and input-reflection ($|S_{11}|$) responses of the theoretically-designed overall transmission-line-based RF filter in Fig. 11 and $|S_{21}|$ responses of the same filter without the input/output lowpass filtering (LPF) sections and of the separate LPF section (design parameters: $Z_1 = 72 \Omega$, $Z_2 = 115 \Omega$, $Z_3 = 145 \Omega$, $Z_{A1} = Z_{A2} = Z_{A3} = 130 \Omega$, $Z_{L1} = Z_{L2} = 112 \Omega$, $Z_S = 145 \Omega$, $f_0 = 1.65$ GHz, $f_1 = 2.3$ GHz, $f_2 = 3.67$ GHz, $f_3 = 6.07$ GHz, and $f_L = 5$ GHz).

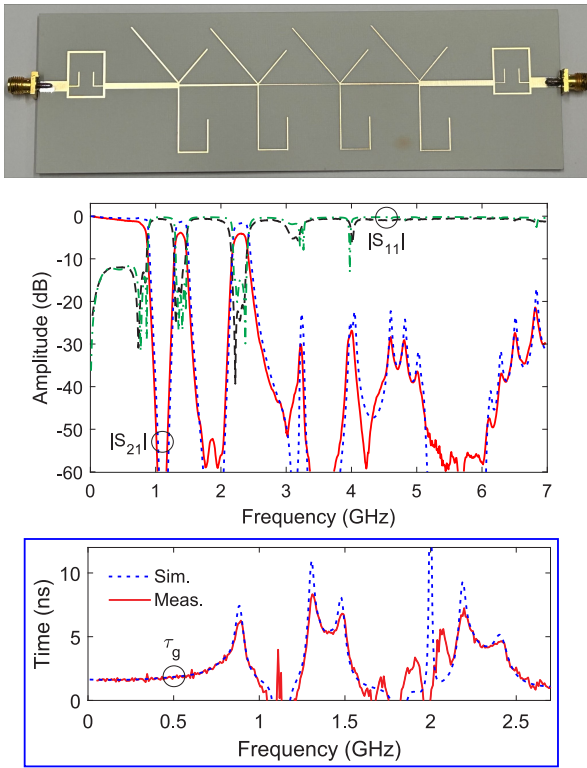


Fig. 13. Photograph (core dimensions: 158.6 mm $\times$ 43.6 mm) and EM-simulated and measured power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the manufactured microstrip prototype of lowpass-dual-passband RF filter.

this filter without the input/output lowpass filtering sections and the separate lowpass filtering section are also represented. As can be seen, the bandwidth and power-attenuation levels of the upper stopband are considerably increased in the overall filter due to the action of the lowpass filtering sections. The presence of spurious narrow-band spikes throughout the entire upper-stopband range is also observed, which is an intrinsic phenomenon to the cascading process as it was studied in [46].

The photograph of the constructed microstrip prototype of lowpass-dual-passband RF filter is shown in Fig. 13. Its EM-simulated and measured power transmission, input-reflection, and in-transmission group-delay responses are also drawn.

Note that if required by the intended application, higher levels of circuit-size reduction can be obtained through the meandering or fractal-shape patterning of the transmission-line segments of the prototype as reported in [47] and [48]. In such case, an additional fine optimization step of the total filter circuit to counteract the extra spurious couplings that may be generated owing to the closer physical spacings resulting among transmission-line segments may be necessary. As can be seen in Fig. 13, a fairly-close agreement between predicted and experimental results is attained. Higher minimum in-band power-insertion-loss levels are visible in the measurements with regard to the simulations, which is attributed to higher-than-expected substrate and radiation losses. Note also that, as design trade-off, such insertion-loss levels are augmented in the overall filter—i.e., with regard to those inherent to the in-series cascade connection of the three triple-open-ended-stub cells—by the extra input/output transversal-signal-interference lowpass-type filtering units that were incorporated to further enlarge the upper-stopband bandwidth. The measured lowpass transmission range has a 3-dB cut-off frequency of 0.85 GHz and a minimum in-band power-matching level of 12 dB. The measured first and second passbands exhibit center frequencies of 1.4 and 2.31 GHz, 3-dB absolute bandwidths equal to 163 and 212 MHz—i.e., 11.6% and 9.2% in relative terms—, minimum in-band power-insertions-loss levels of 3.9 and 4.1 dB, and in-band power-matching levels higher than 11.5 and 20.9 dB, respectively. The measured upper-stopband range extends from 2.58 to 6.79 GHz for a minimum attenuation level of 26.9 dB. Note also that, due to the substrate loss, the effect of spurious spikes in the measured upper stopband when compared to the theoretical response depicted in Fig. 12 is largely reduced. The measured maximum in-band group-delay variations for the lowpass-type band and the first and second passbands are 3.9, 2.9, and 2.5 ns, respectively.

A qualitative comparison of the manufactured lowpass-dual-passband filter prototype in Fig. 13 with other related prior-art filters is provided in Table II. As can be seen, the co-designed filter developed in this work exhibits all the desired features in terms of TZ generation at both sides of all transmission bands, extended upper stopband with extra TZ creation, and low circuit complexity that results in ease of scalability to higher-order realizations with increased number of passbands. Besides, it is the only one whose design is based on a generalized frequency transformation—i.e., Sub-Type II-A in this case.

B. Design II: Inverterless Lumped-Element Filter

As the second example, a third-order lowpass-dual-passband RF filter in lumped-element technology for the VHF/UHF range has been designed. In this case, the generalized frequency transformation Sub-Type II-B was employed to derive the converted co-designed filter network. Afterwards, circuit transformations were applied to this circuit network to remove the admittance inverters. In this manner, the negative effects associated to the frequency dependency of the admittance inverters in a lumped-element realization in terms of distortion of the overall filtering response are avoided. It also results in an implementation with minimum number of lumped elements.

TABLE II
COMPARISON WITH PRIOR-ART RF LOWPASS-SINGLE/MULTI-PASSBAND FILTERS

	[18]: Fig. 3	[19]: Fig. 5	[21]: Fig. 7	[22]: Fig. 9	[24]: Fig. 2	[25]: Fig. 6	This work: Fig. 13
Circuit structure and technology	Microstrip filter bank	Microstrip in-parallel sets of resonators	Microstrip metamaterial-inspired resonant cells	Microstrip double-shunted open-ended-line stages	Microstrip stepped-impedance transversal-signal-interference multi-band sections	Microstrip stepped-impedance sections with etched slots	Microstrip multi-open-ended-stub stages with input/output lowpass-type cells
Type of response (frequencies ^(†) in GHz)	Lowpass-single-passband (1.5/2.4)	Lowpass-dual-passband (0.66/1.57/2.45)	Lowpass-dual-passband (1.05/1.9/3)	Lowpass-single-passband (1.5/2.42)	Lowpass-single-passband (1/2.25)	Lowpass-single-passband (1 ^(*) /4.75)	Lowpass-dual-passband (0.85/1.4/2.31)
Multiple TZs	✓	✓	✓	✓	✓	×	✓
Extended stopband	✓	✓	×	×	✓	×	✓
Ease of scalability	Medium	Medium	Unclear	High	Medium	High	High

(†): 3-dB cut-off frequency for lowpass-type band and center frequencies for passbands; (*): specified value at ripple-referred level.

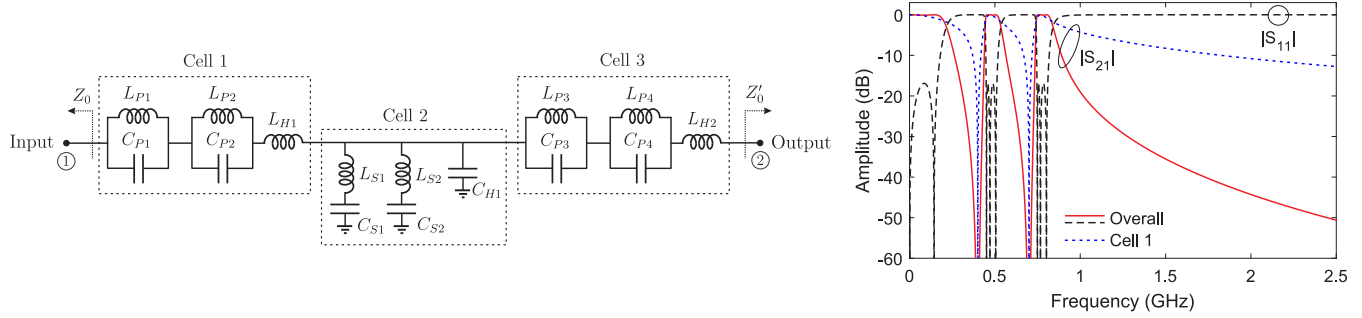


Fig. 14. Circuit detail and power transmission ($|S_{21}|$) and input-reflection ($|S_{11}|$) responses of the theoretically-designed overall lumped-element-based RF filter along with the $|S_{21}|$ response of its multi-resonant cell I—the theoretical/exact values for the design parameters are provided in Table III.

TABLE III
LUMPED-ELEMENT VALUES OF DESIGN II

Element*	Value
$L_{P1} = \left(\frac{J_{C1} J_A}{J_B J_1} \right)^2 \left(\frac{\beta_1}{2\pi f_1^2 Y_0} \right)$	13.816 nH
$C_{P1} = \left(\frac{J_1 J_B}{J_{C1} J_A} \right)^2 \left(\frac{Y_0}{2\pi \beta_1} \right)$	11.459 pF
$L_{P2} = \left(\frac{J_{C2} J_A}{J_B J_1} \right)^2 \left(\frac{\beta_2}{2\pi f_2^2 Y_0} \right)$	4.511 nH
$C_{P2} = \left(\frac{J_B J_1}{J_{C2} J_A} \right)^2 \left(\frac{Y_0}{2\pi \beta_2} \right)$	11.459 pF
$L_{H1} = \left(\frac{J_A}{J_B J_1} \right)^2 \left(\frac{Y_0}{2\pi \alpha} \right)$	27.631 nH
$L_{S1} = \left(\frac{J_B J_2}{J_1 J_{C1} J_A} \right)^2 \left(\frac{Y_0}{2\pi \beta_1} \right)$	25.179 nH
$C_{S1} = \left(\frac{J_1 J_{C1} J_A}{J_B J_2} \right)^2 \left(\frac{\beta_1}{2\pi f_1^2 Y_0} \right)$	6.288 pF
$L_{S2} = \left(\frac{J_B J_2}{J_1 J_{C2} J_A} \right)^2 \left(\frac{Y_0}{2\pi \beta_2} \right)$	25.179 nH
$C_{S2} = \left(\frac{J_1 J_{C2} J_A}{J_B J_2} \right)^2 \left(\frac{\beta_2}{2\pi f_2^2 Y_0} \right)$	2.053 pF
$C_{H1} = \left(\frac{J_A J_1}{J_B J_2} \right)^2 \left(\frac{Y_0}{2\pi \alpha} \right)$	12.575 pF
$L_{P3} = \left(\frac{J_2 J_{C1} J_A}{J_B J_1 J_3} \right)^2 \left(\frac{\beta_1}{2\pi f_1^2 Y_0} \right)$	13.816 nH
$C_{P3} = \left(\frac{J_B J_1 J_3}{J_2 J_{C1} J_A} \right)^2 \left(\frac{Y_0}{2\pi \beta_1} \right)$	11.459 pF
$L_{P4} = \left(\frac{J_2 J_{C2} J_A}{J_B J_1 J_3} \right)^2 \left(\frac{\beta_2}{2\pi f_2^2 Y_0} \right)$	4.511 nH
$C_{P4} = \left(\frac{J_B J_1 J_3}{J_2 J_{C2} J_A} \right)^2 \left(\frac{Y_0}{2\pi \beta_2} \right)$	11.459 pF
$L_{H2} = \left(\frac{J_A J_2}{J_B J_1 J_3} \right)^2 \left(\frac{Y_0}{2\pi \alpha} \right)$	27.631 nH
$Z'_0 = \left(\frac{J_2 J_A}{J_1 J_3} \right)^2 Z_0$	50 Ω (output port)

*Design parameters: $f_1 = 0.4$ GHz, $f_2 = 0.7$ GHz, $\alpha = 0.2$ GHz, $\beta_1 = 0.1$ GHz, $\beta_2 = 0.1$ GHz, $Y_0 = 1/Z_0$ with $Z_0 = 50 \Omega$, $J_1 = J_4 = 1.2Y_0$, $J_2 = J_3 = 1.35Y_0$, $J_A = J_B = Y_0$, and $J_{C1} = J_{C2} = 1.3Y_0$.

The circuit detail of the inverterless lumped-element third-order lowpass-dual-passband filter is shown in Fig. 14. The formulas for its inductance and capacitance elements, along with their exact values for the selected design, are given in Table III. Steps 1)—but with the design equations (13)–(18)—, 2) and 5) described in Section IV-A were followed in its design process. Note that the realization of more-complex filtering transfer functions, such as those also including a highpass term or more passbands, would lead to constituent lumped-element cells with more inductive and capacitive elements—to be derived in each case using an analogous methodology. The power transmission and input-reflection responses of this theoretically-designed third-order lowpass-dual-passband filter, along with the power transmission response of its first multi-resonance cell for selectivity-comparison purposes, are also depicted in Fig. 14. The in-band ripple level for all transmission bands is 0.1 dB, whereas the ripple-level-referred cut-off frequency of the lowpass-type transmission range is 165 MHz. The center frequencies of the first and second passbands are 470 and 770 MHz, respectively, and their ripple-level-referred absolute bandwidths are 62 MHz.

V. CONCLUSION

Various types of generalized frequency transformations for co-designed RF filtering devices have been presented. They allow the direct realization of arbitrary filtering responses with composite-lowpass-single/multi-passband/highpass terms. For these frequency transformations, the resulting co-designed filtering networks after their application over the normalized equivalent lowpass filter prototype in terms of lumped-element models have been detailed. Besides, tech-

niques for sharper-rejection/increased-selectivity RF filter design based on the inclusion of additional coupling paths in different ways to create extra stopband TZs have been described. The usefulness of such generalized frequency transformations also extends to RF multi-functional filtering components, as it has been validated by means of theoretical design examples consisting of an input-reflectionless filter and two-way filtering power dividers with equal and unequal power-splitting factors. Moreover, two examples of RF filters designed in distributed-element and inverterless lumped-element circuits have been presented, along with the measured results of a manufactured proof-of-concept microstrip prototype corresponding to the first RF filter design for experimental-demonstration purposes.

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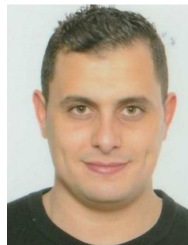
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